

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

DIGITAL SYSTEM DESIGN

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions carry marks as indicated.
- (2) All questions are compulsory.
- (3) Assume suitable data wherever necessary.
- (4) Illustrate your answers with the help of neat sketches wherever necessary.

1. Solve using K Map :

$$f(A, B, C, D) = \Sigma(1, 2, 3, 6, 7, 8, 9, 11, 14, 15)$$

- (I) List all essential and non-essential prime implicants.
- (II) Simplify the function f to get minimum-cost circuit.
- (III) Design gate level digital circuit and write Verilog code for the same using primitives. 10(CO1,4)

2. (a) Demonstrate event queue in Verilog with example. 6(CO2)(b) Compare blocking and non-blocking statements in Verilog. 4(CO2)

3. (a) Design a digital circuit which has three input lines (a, b, clk), where a and b are control inputs and clk is clock input signal with 20 KHz frequency, it also has one output (Y). Depending on the values of input control lines, output signal frequency should change as follows :

Input a	Input b	Output (y) signal frequency
1	1	5 KHz
1	0	10 KHz
0	1	2.5 KHz

Write the Verilog code for above digital circuit.

10(CO4)

4. Design state diagram and write Verilog code for the Sequence detector FSM which detects overlapping sequence "101010". 10(CO4)
5. Design the structure of 2 input LUT FPGA with implementation of Boolean function $(X_1X_3\bar{X}_6 + X_1X_4X_5\bar{X}_6 + X_2X_3X_7 + X_2X_4X_5X_7)$ 10(CO1,5)
6. Classify semiconductor logic families and compare them with respect to the following parameters :
- (i) Propagation delay.
 - (ii) Power dissipation.
 - (iii) Noise margin.
 - (iv) Fan in and fan out.
 - (v) Cost. 10(CO3)

